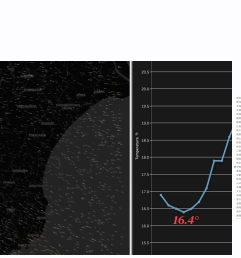
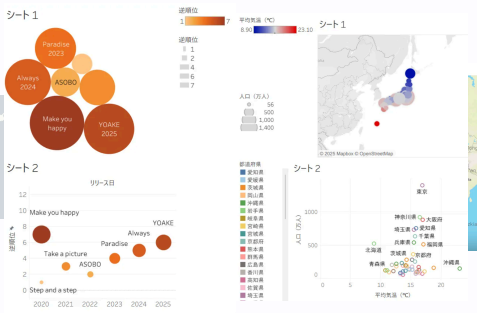
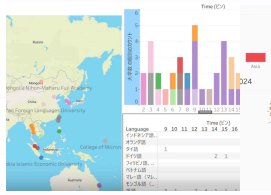
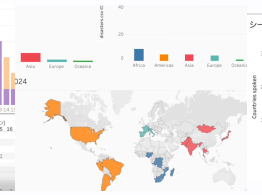
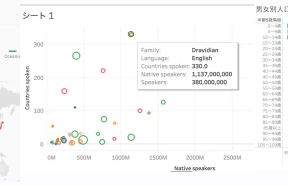
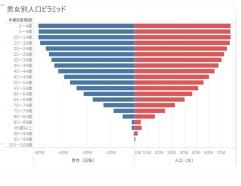
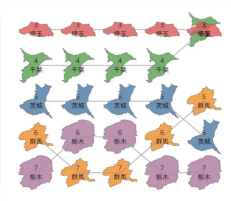
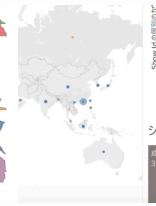
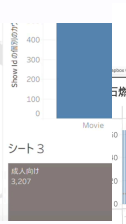
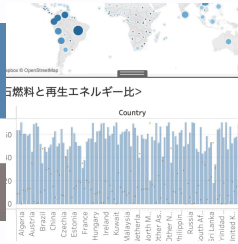
Gallery walk



0598

Padlet link

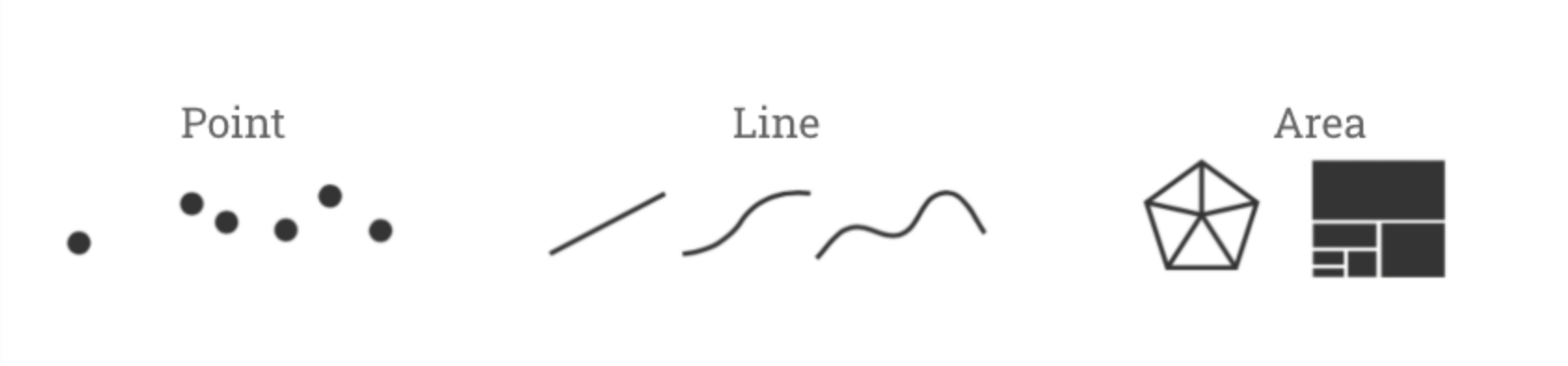
Language of Data Visualization

Marks and Channels



Marks ・　マーク

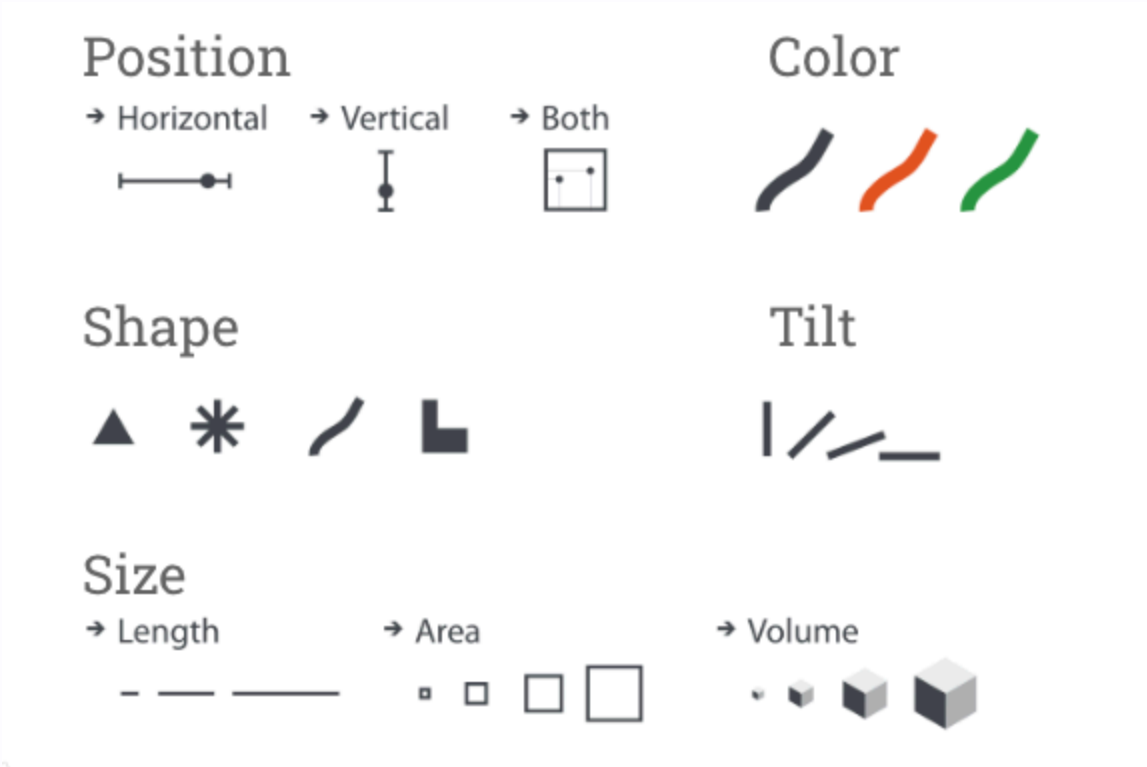
The shapes that represent data.
データを表す形

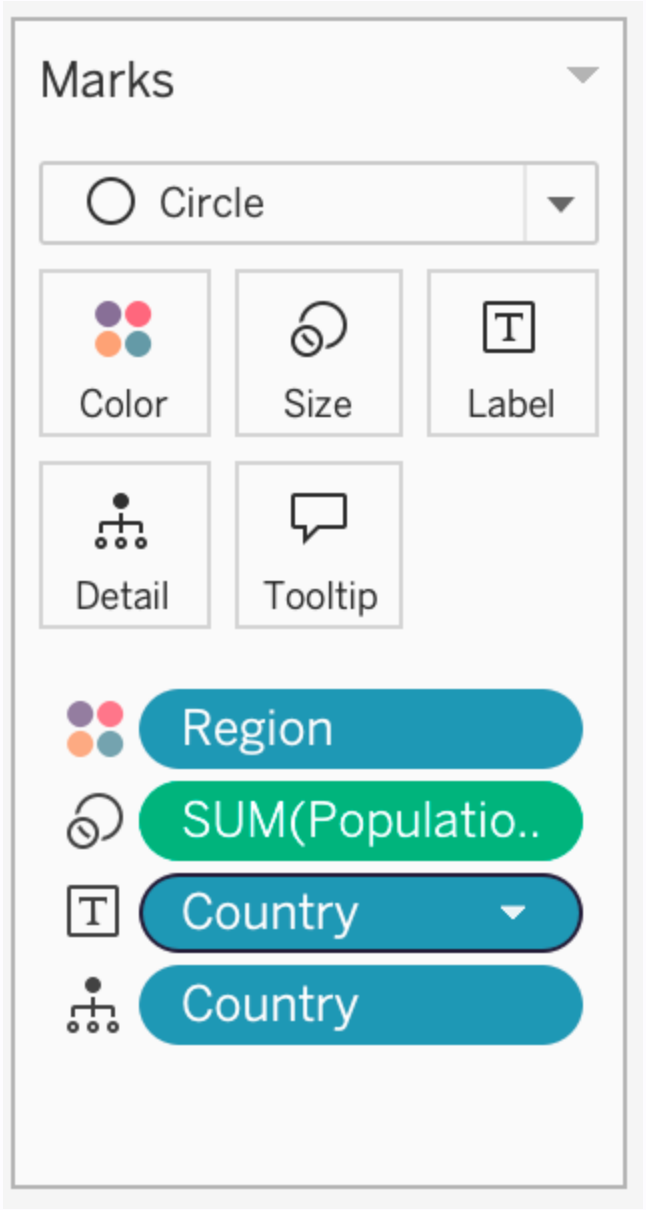


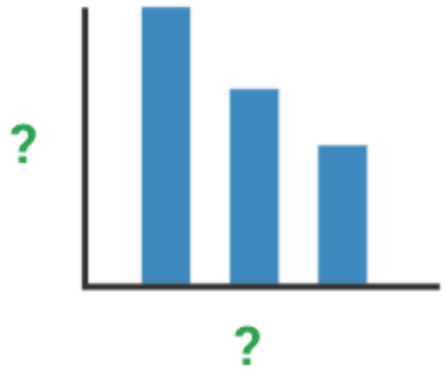
Graphics from Munzner. Visualization Analysis and Design (2015)

Channels ・ チャネル

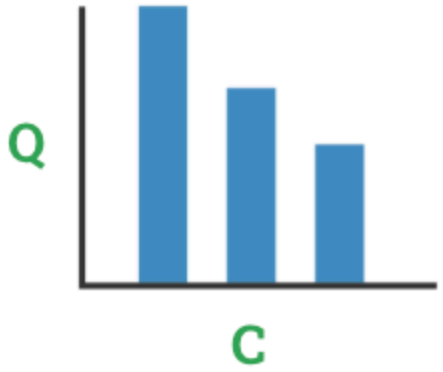
Ways to vary the appearance of marks
マークの見た目を変える方法



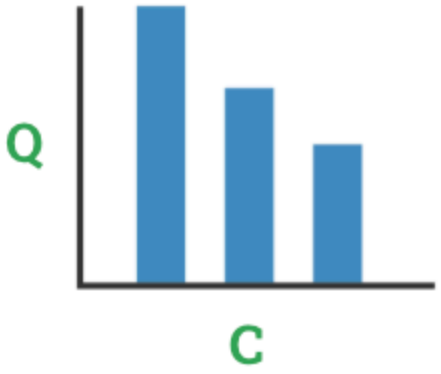

Marks
Circle
Color
Size
Label
Detail
Tooltip
Region
SUM(Populatio..
Country
Country



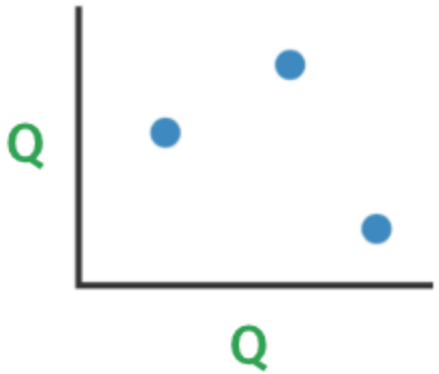
Mark: ?



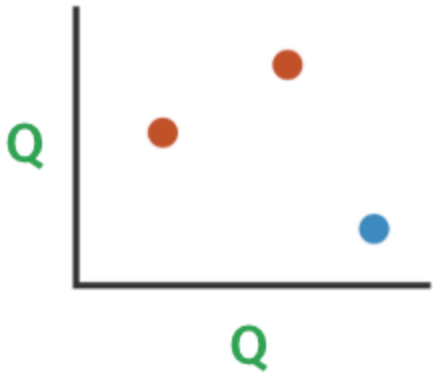
Mark: **line**



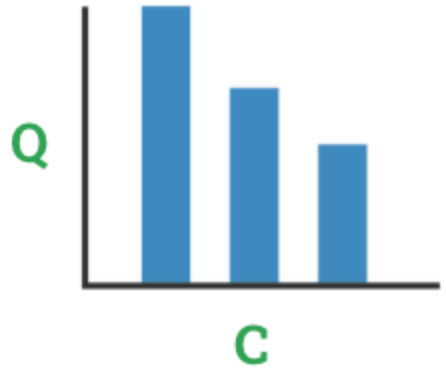
Mark: **line**



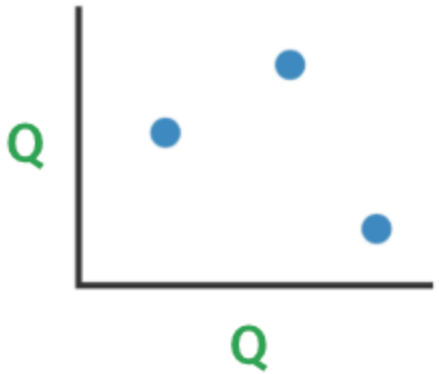
Mark: **point**



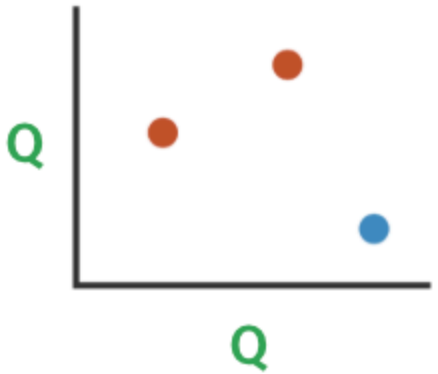
Mark: **point**
Channels:
Color: **C**



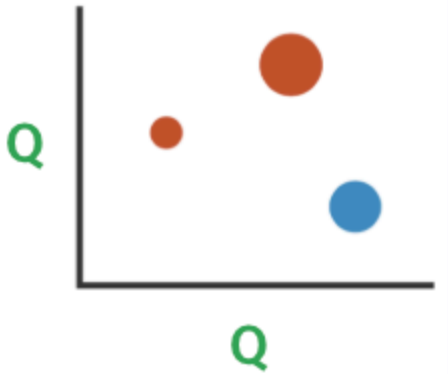
Mark: **line**



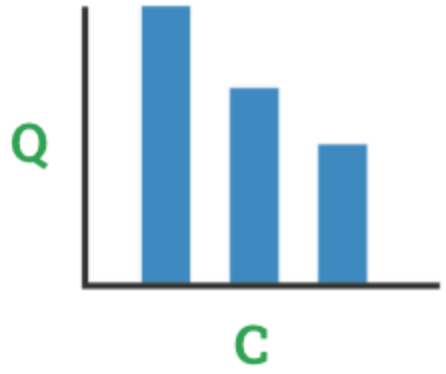
Mark: **point**



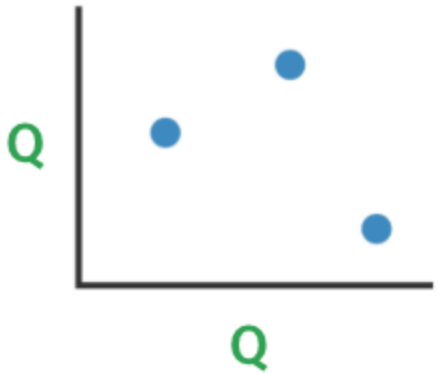
Mark: **point**
Channels:
Color: **C**



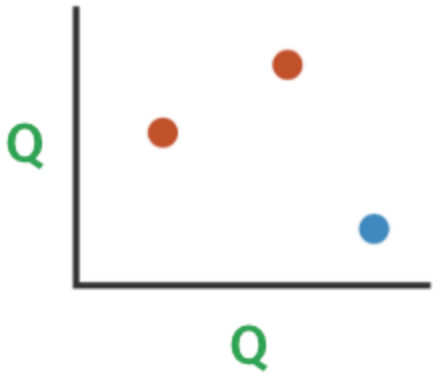
Mark: **point**
Channels: **?**



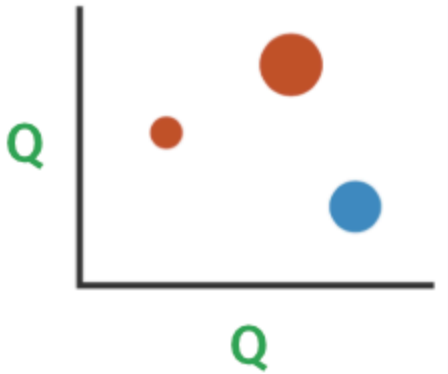
Mark: **line**



Mark: **point**



Mark: **point**
Channels:
Color: **C**



Mark: **point**
Channels:
Color: **C**
Size: **Q**

Position

→ Horizontal → Vertical → Both



Color



Shape



Tilt



Size

→ Length → Area → Volume



Position

→ Horizontal → Vertical → Both



Color



Categorical



Shape



Tilt



×

Ordinal



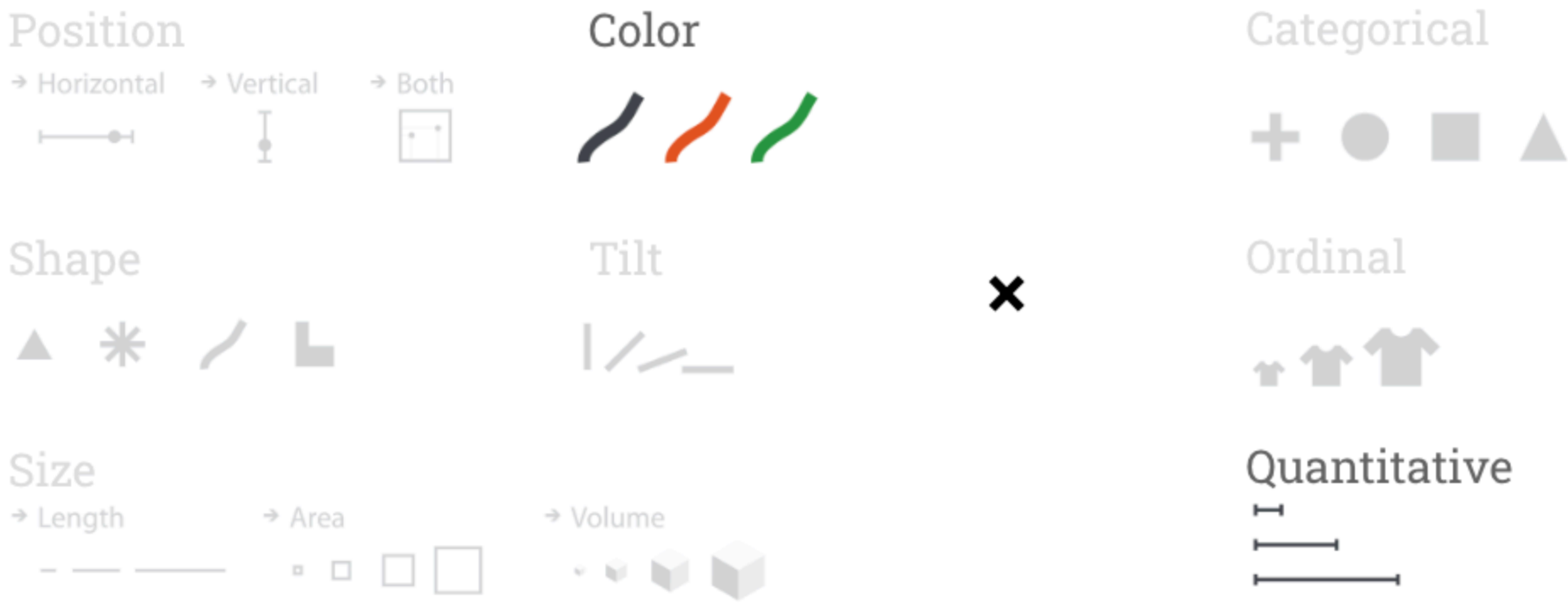
Size

→ Length → Area → Volume



Quantitative




Position
Horizontal
Vertical
Both
Color
Categorical
Shape
Tilt
Ordinal
Size
Length
Area
Volume
Quantitative

Filters

Marks

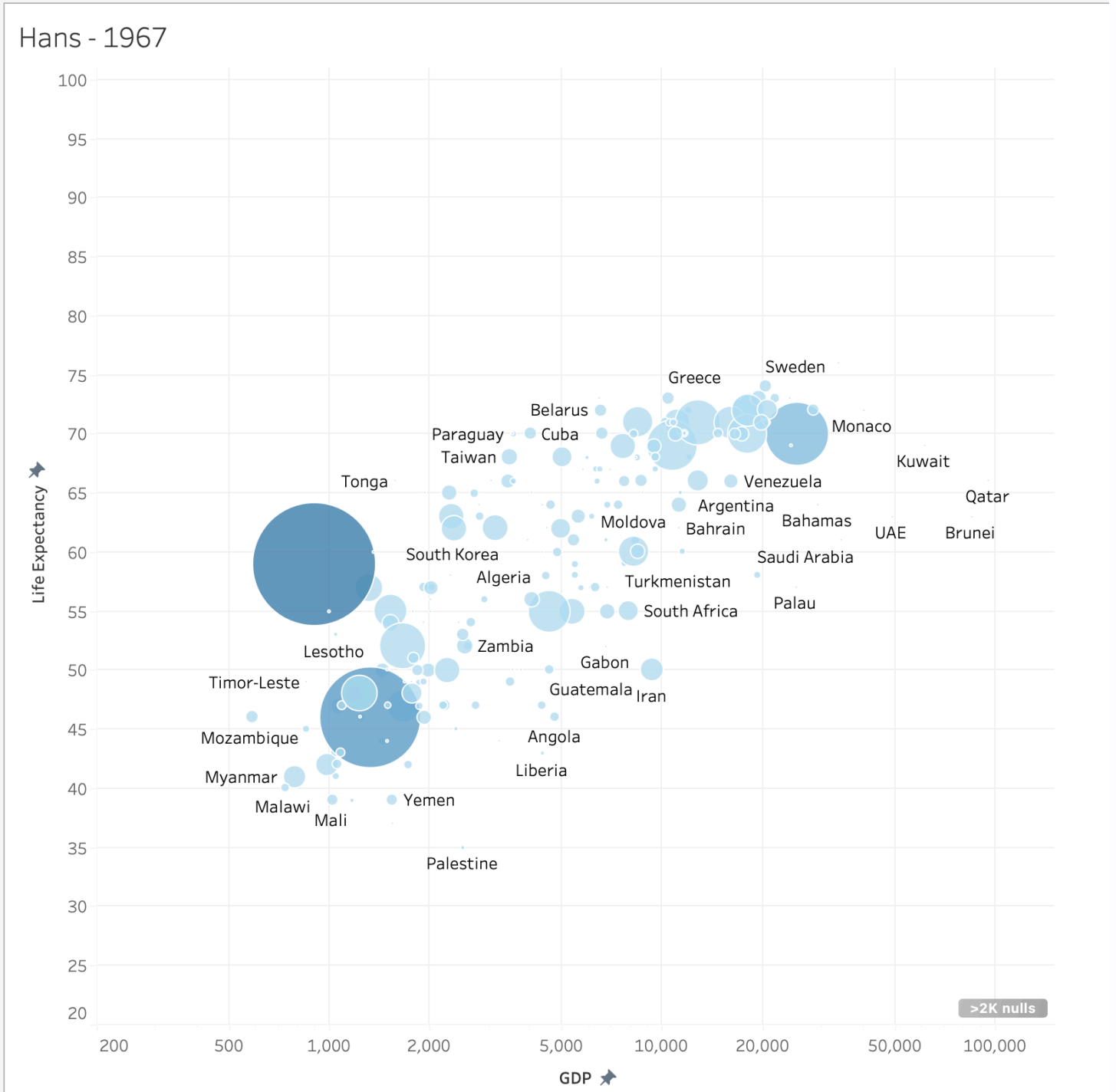
Circle

Color | Size | Label | Detail | Tooltip

SUM(Populatio..
SUM(Populatio..
Country
Country

SUM(Population)

651 999M



Position

→ Horizontal → Vertical → Both

Color



Categorical



Shape

Tilt

×

Ordinal

Size

→ Length → Area → Volume

Quantitative

Filters

Marks

Circle

Color | Size | Label

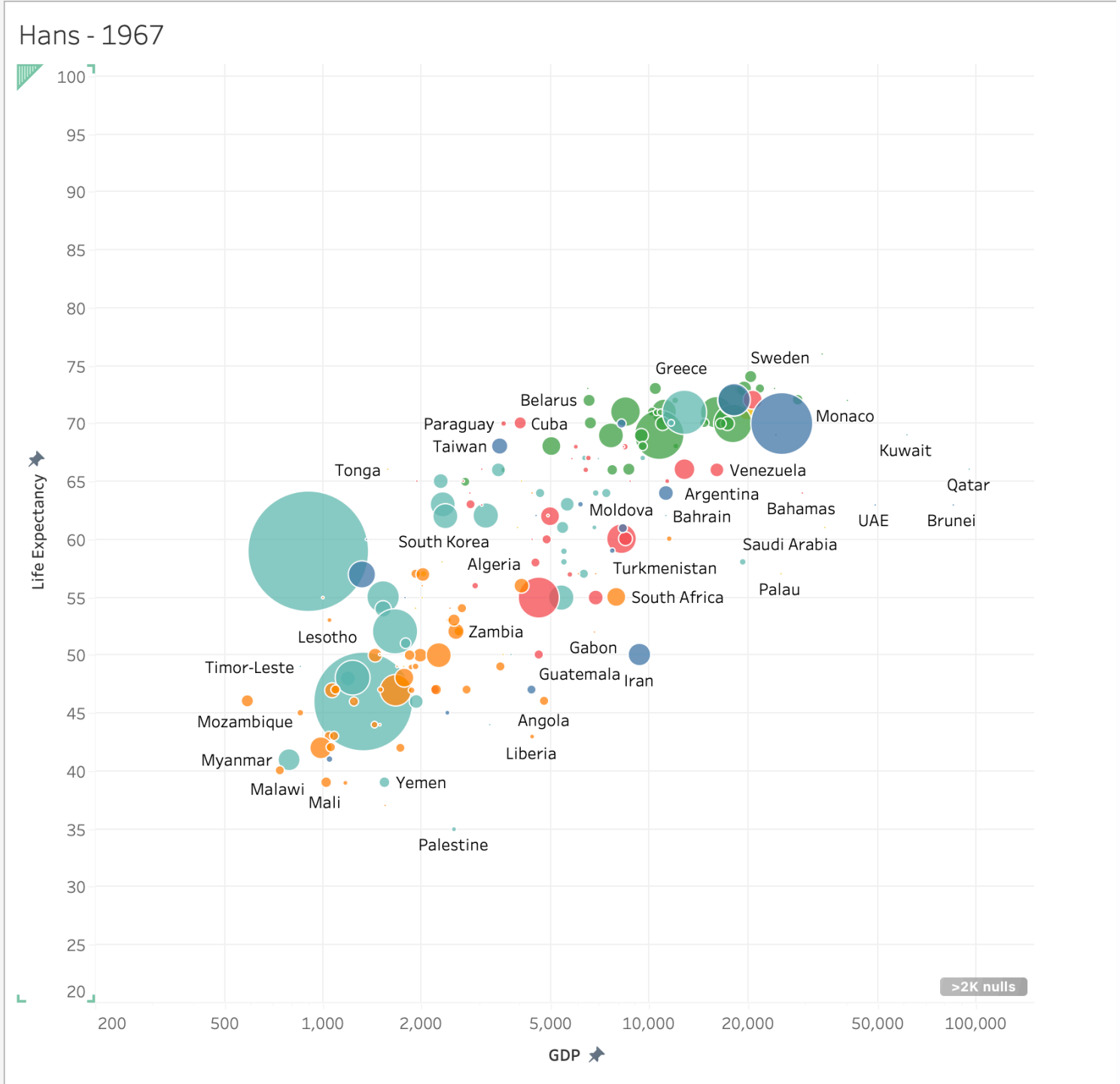
Detail | Tooltip

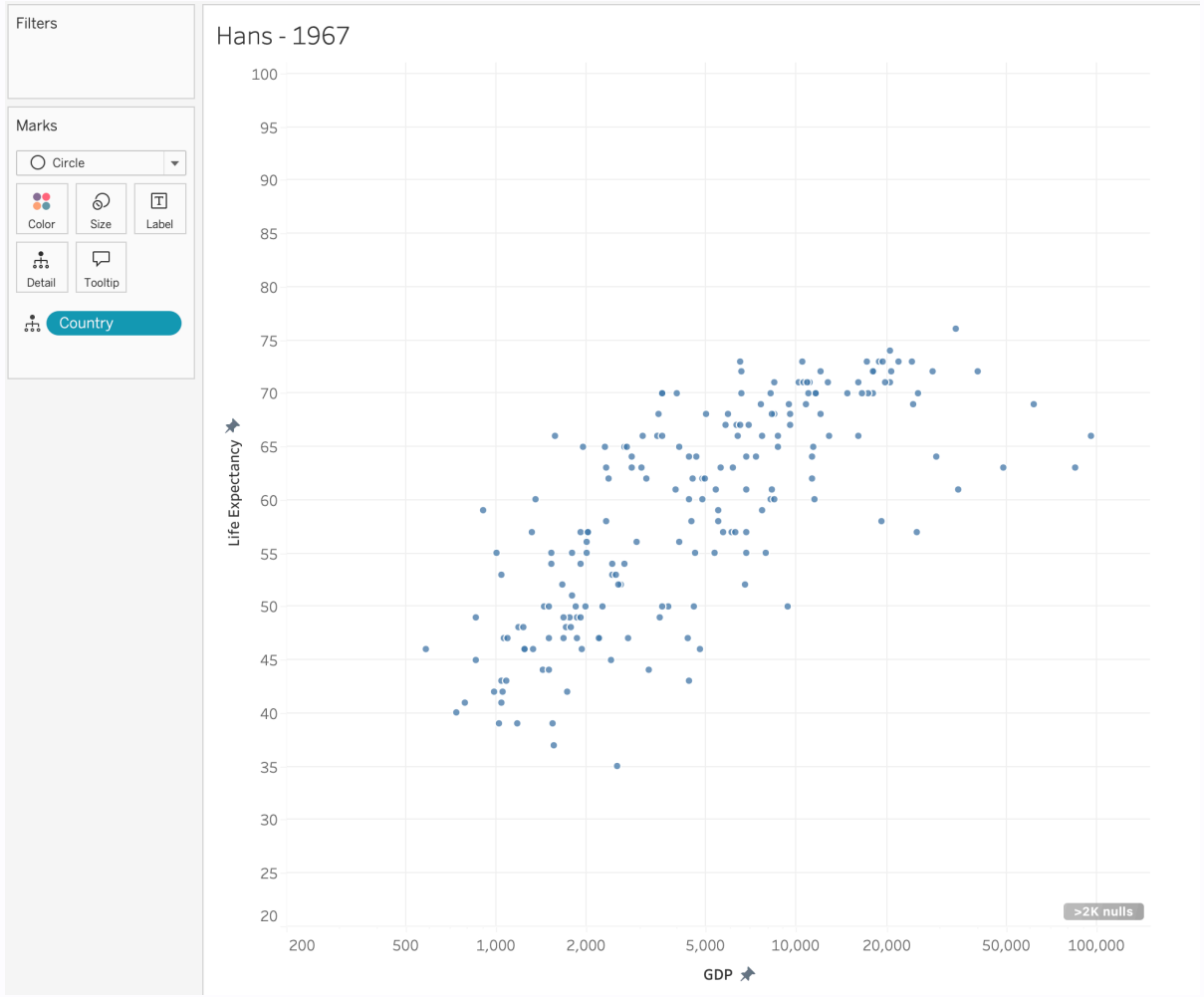
Region

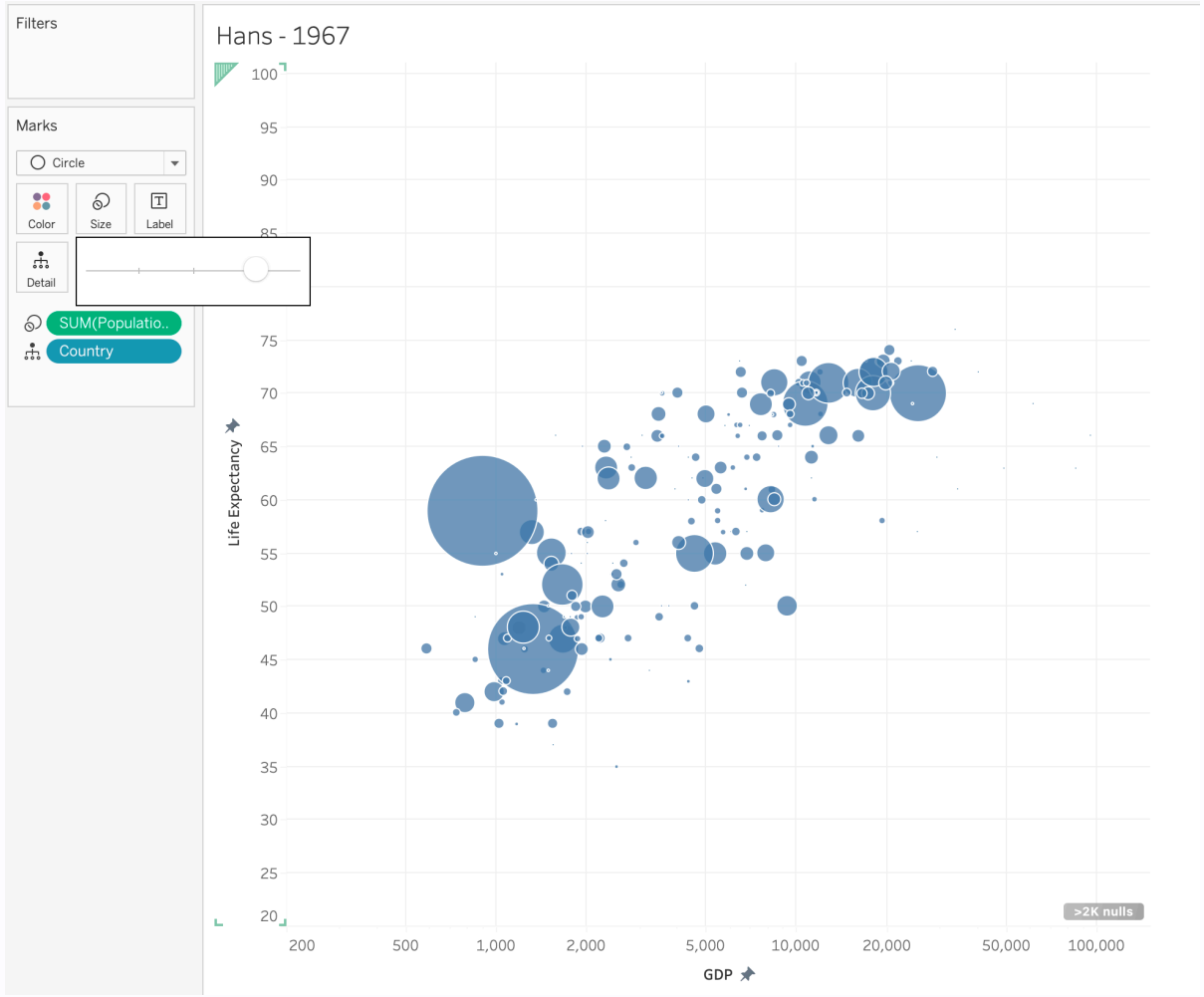
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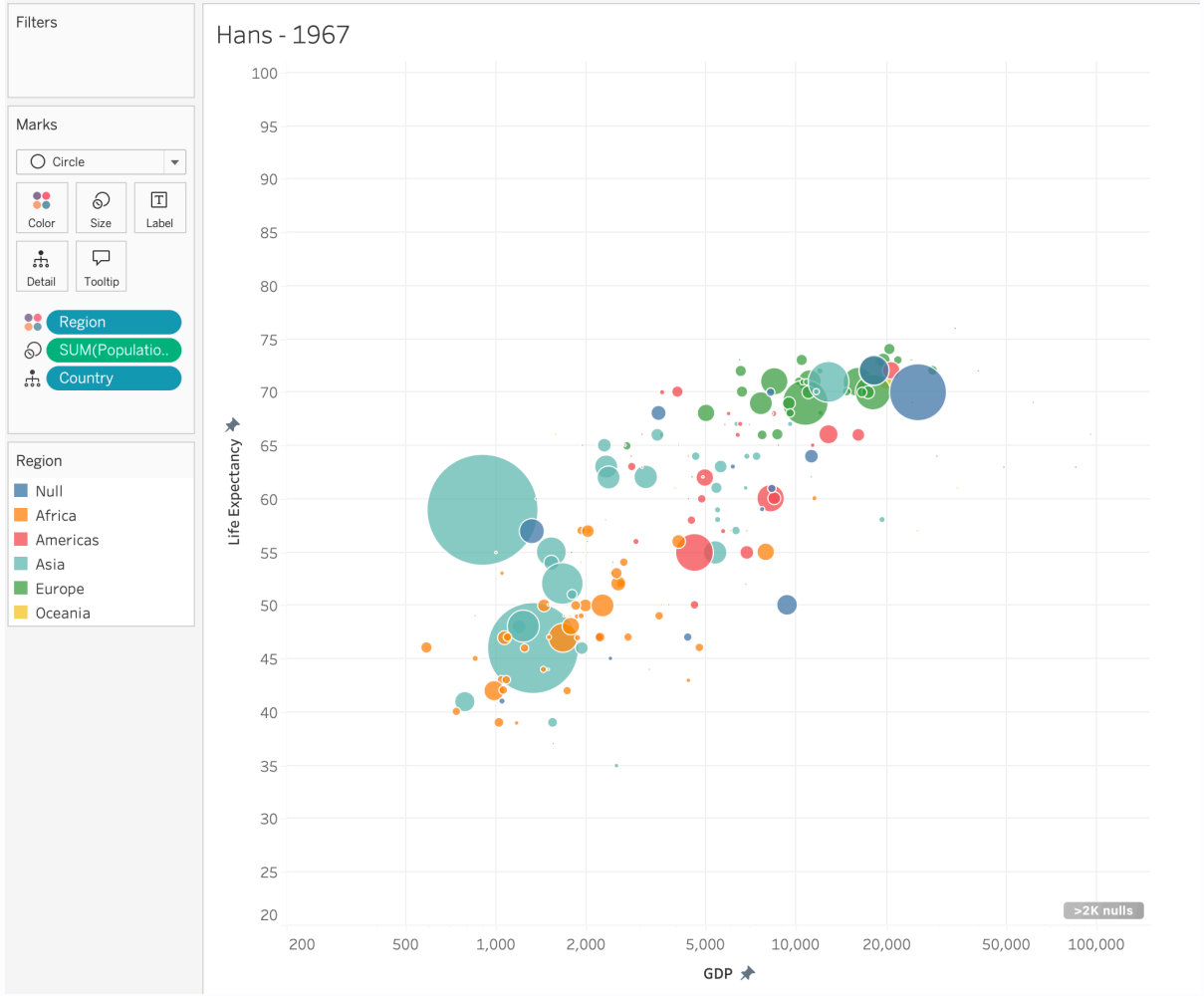
Country

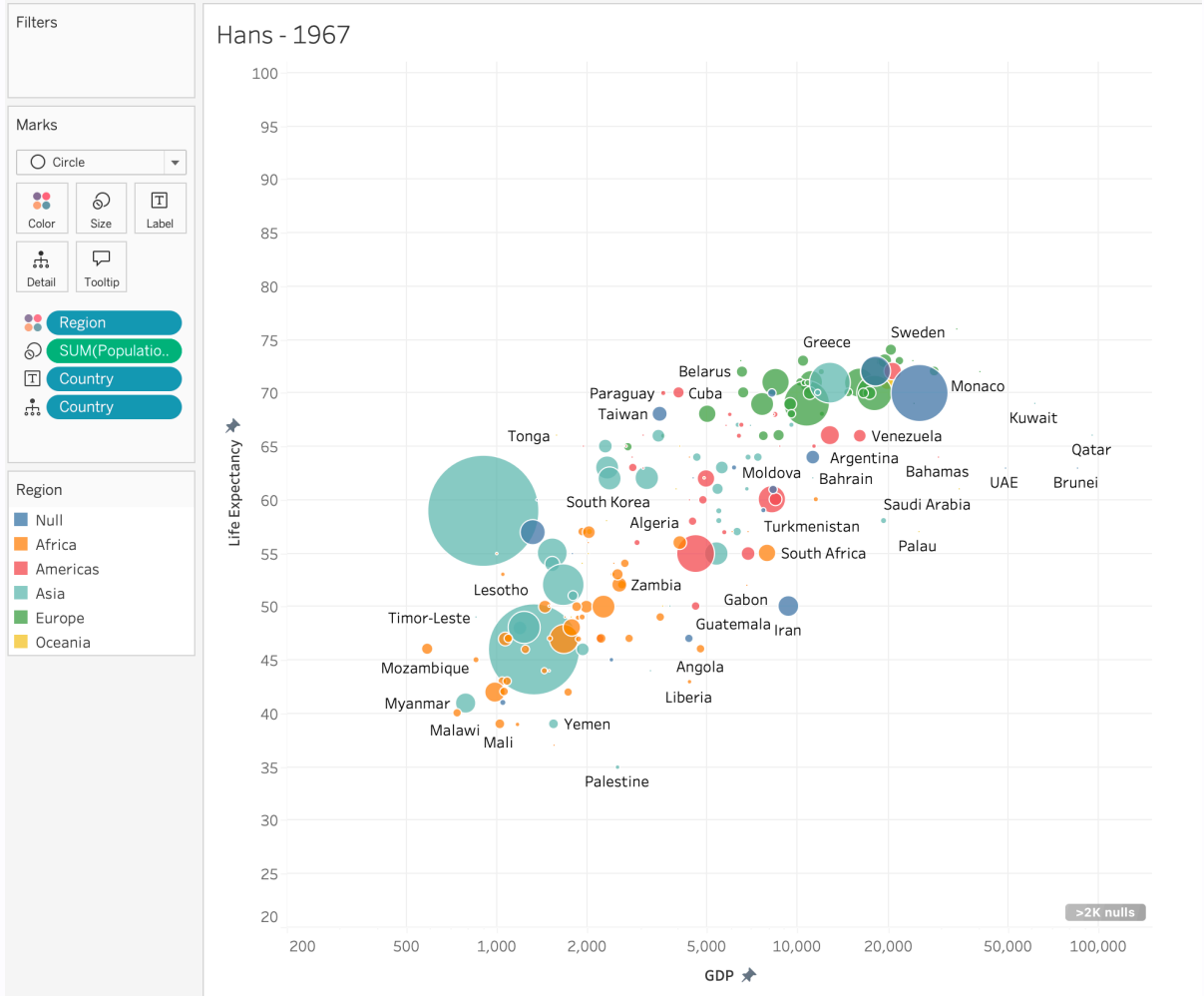
Country

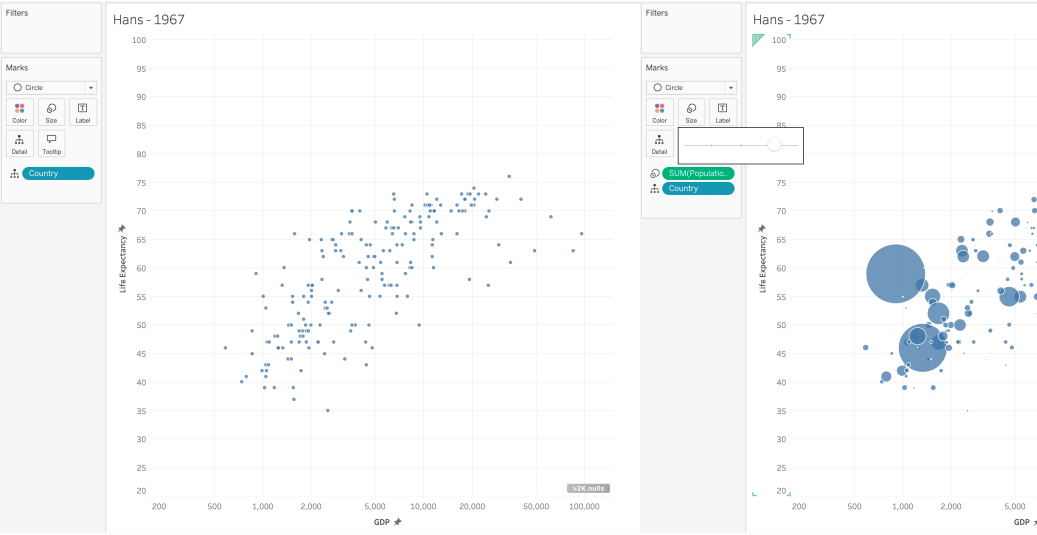


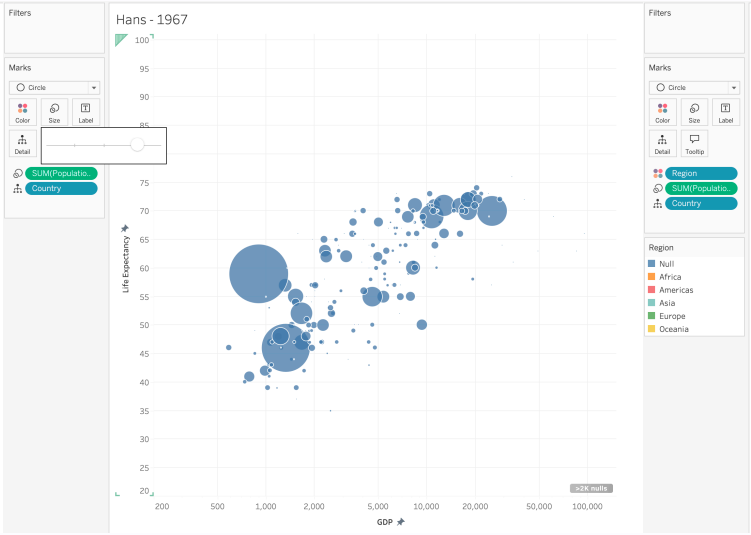

Filters
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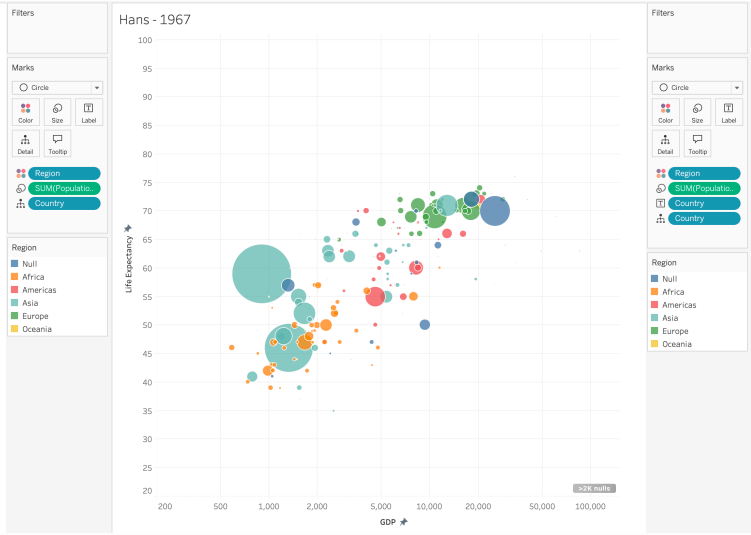

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Filters
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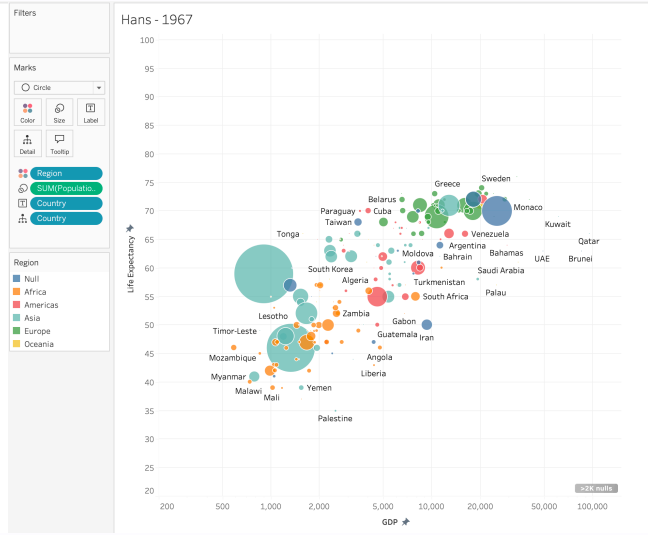

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Monaco
Paraguay
Cuba
Taiwan
Kuwait
Tonga
Venezuela
Qatar
Argentina
Moldova
Bahrain
Bahamas
UAE
Brunei
South Korea
Saudi Arabia
Algeria
Turkmenistan
Palau
South Africa
Zambia
Lesotho
Gabon
Timor-Leste
Guatemala
Iran
Mozambique
Angola
Liberia
Myanmar
Malawi
Yemen
Mali
Palestine
>2K nulls


Filters
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Hans - 1967
Life Expectancy
GDP
>2K nulls


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Hans - 1967
Life Expectancy
GDP
>2K nulls


Filters
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Americas
Asia
Europe
Oceania
Hans - 1967
Life Expectancy
GDP
>2K nulls

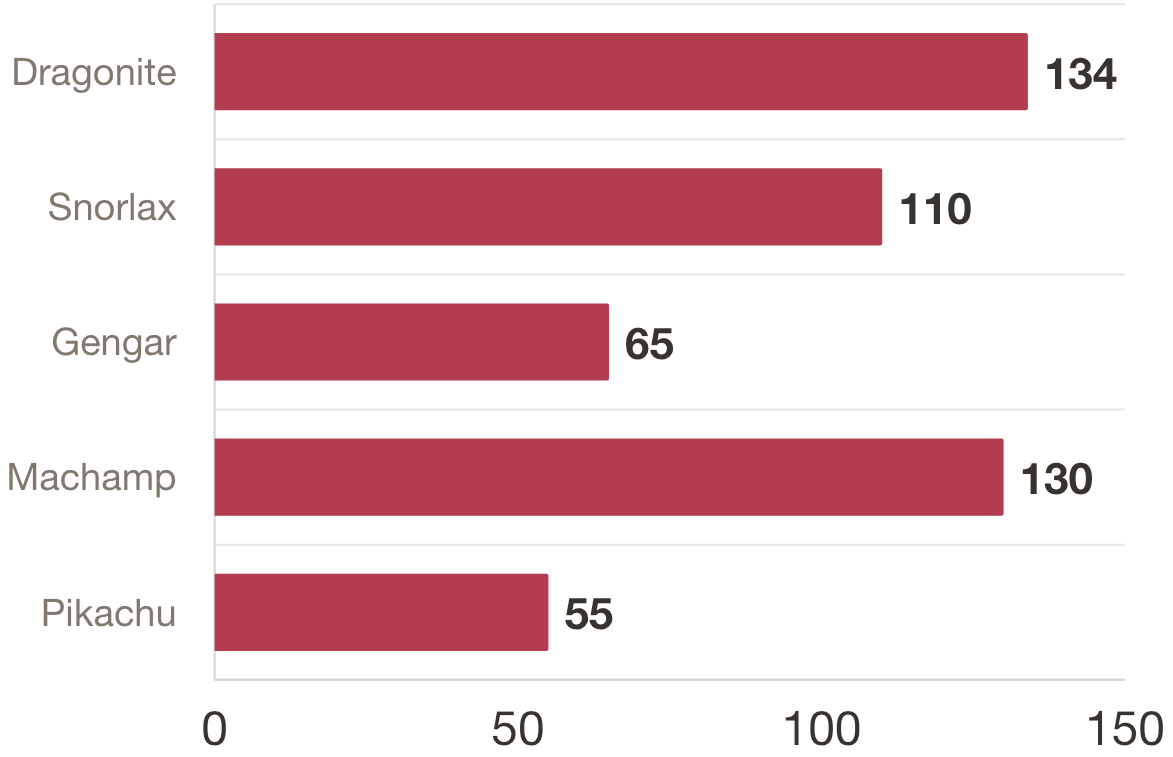
Filters
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SUM(Populatio..
Country
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Africa
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Oceania
Hans - 1967
Life Expectancy
GDP
>2K nulls


Hans - 1967
Life Expectancy
GDP
>2K nulls
Greece
Sweden
Belarus
Monaco
Paraguay
Cuba
Taiwan
Kuwait
Tonga
Venezuela
Argentina
Qatar
Moldova
Bahrain
Bahamas
UAE
Brunei
South Korea
Saudi Arabia
Algeria
Turkmenistan
Palau
South Africa
Lesotho
Zambia
Gabon
Timor-Leste
Guatemala
Iran
Mozambique
Angola
Liberia
Myanmar
Malawi
Yemen
Mali
Palestine

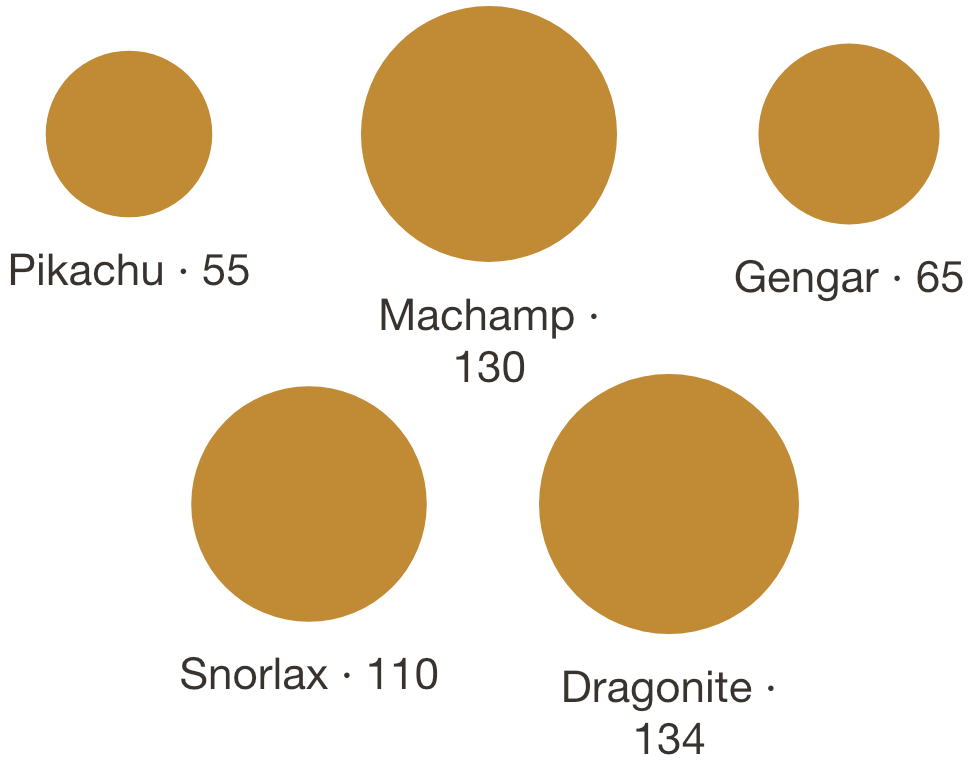
Which comparison is easier?

Same Pokémon. Same Attack values.

POSITION + LENGTH



AREA

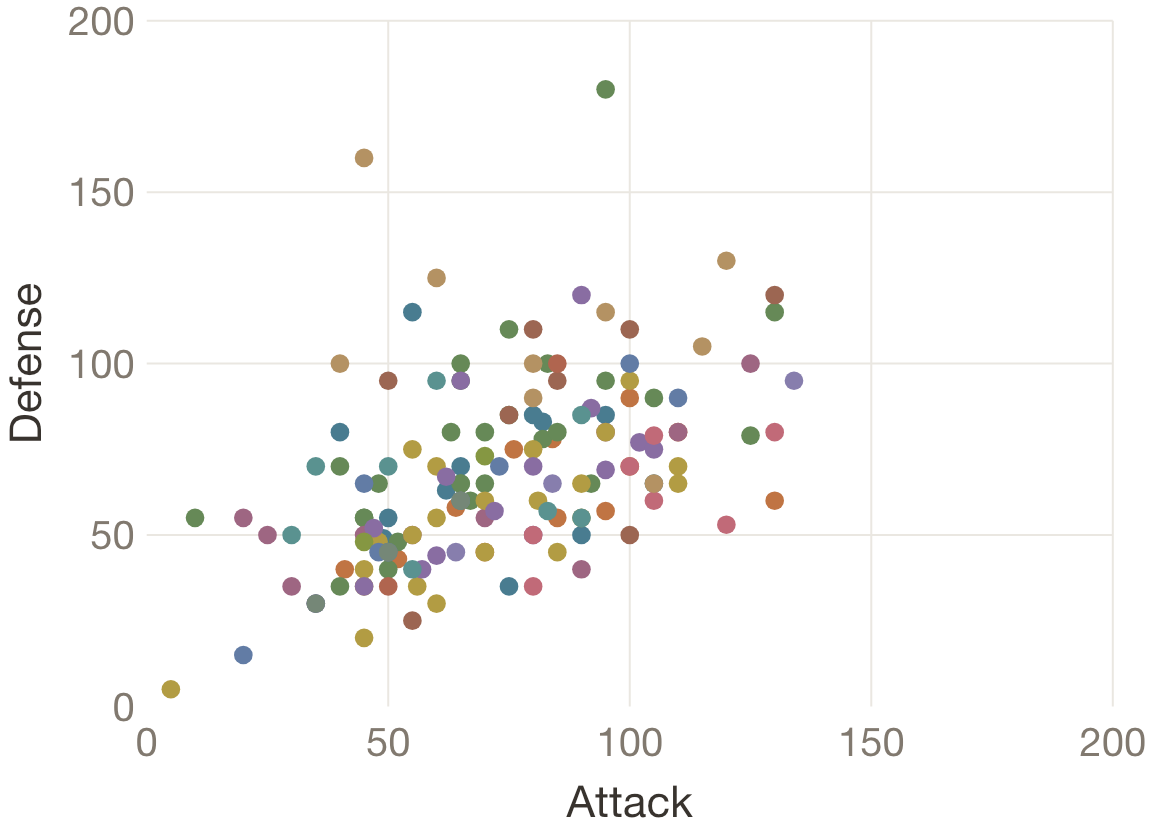


5 selected species · base Attack

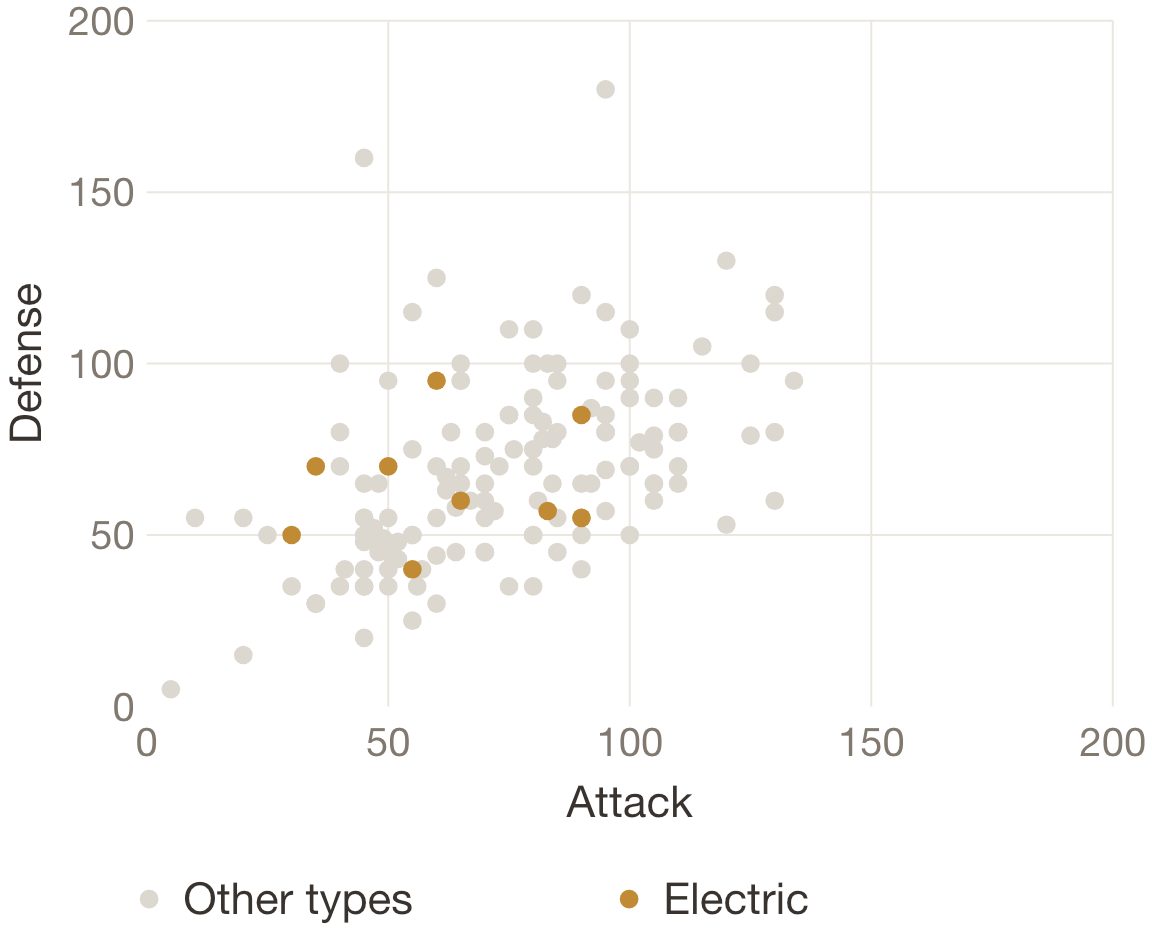
What catches your eye first?

Color everything — or direct attention?

ALL TYPES



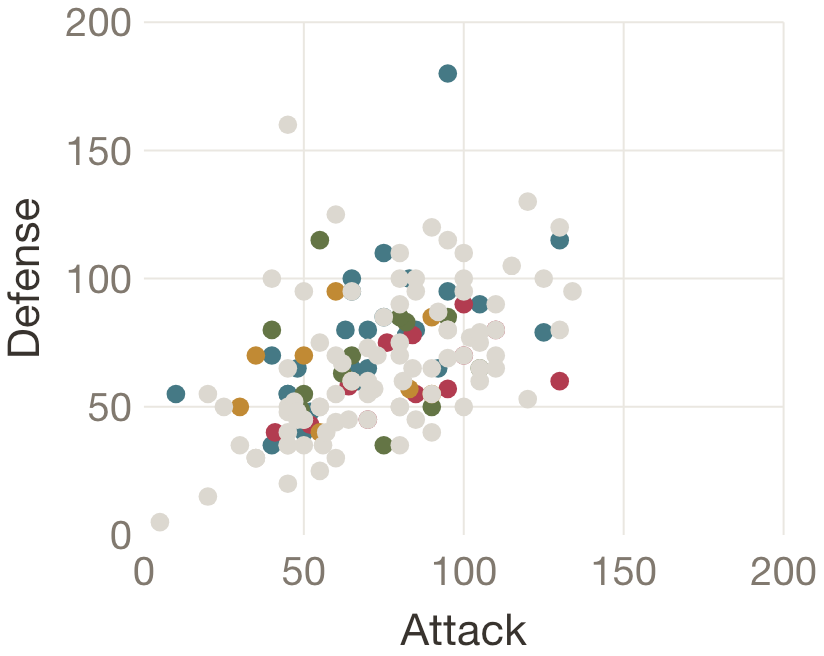
ELECTRIC IN FOCUS



Color needs a job.

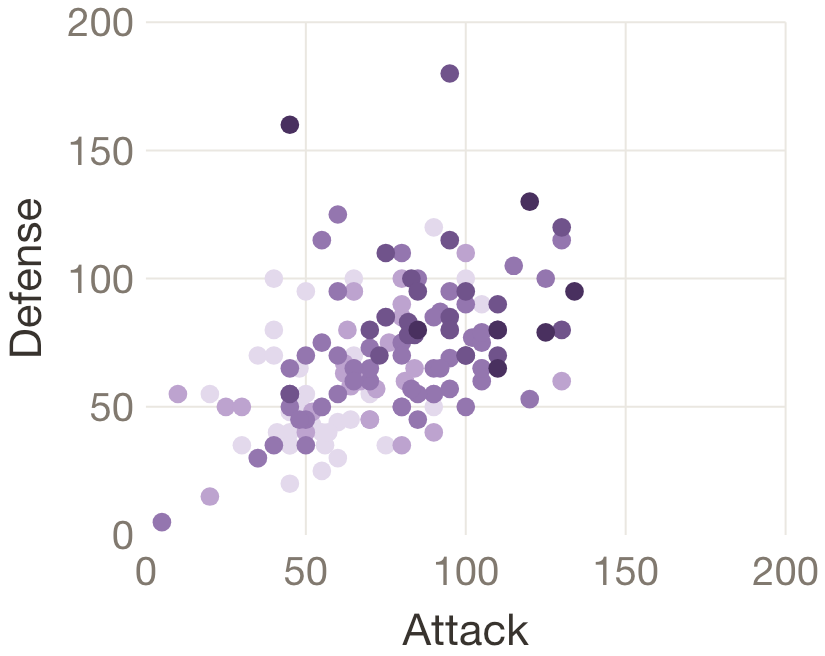
Different kinds of data need different color logic.

TYPE



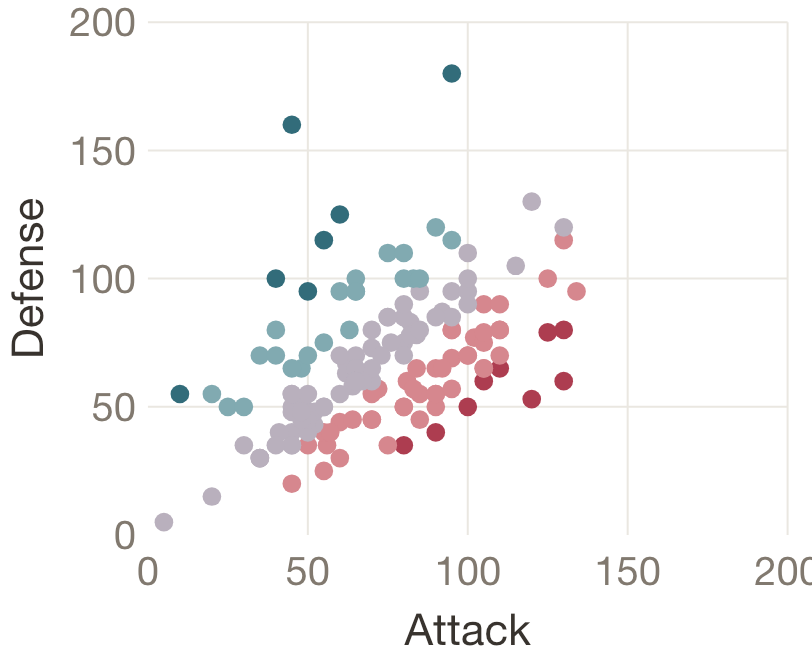
- Water
- Fire
- Grass
- Electric
- Other

WEIGHT



- 0–<10 kg
- 10–<30 kg
- 30–<70 kg
- 70–<150 kg
- 150+ kg

DEFENSE – ATTACK



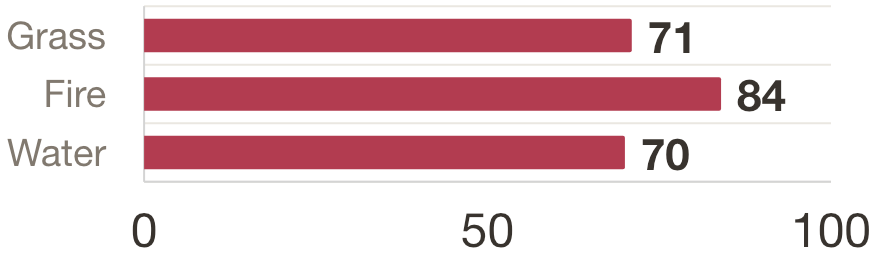
- ≤–41
- –40…–11
- –10…10
- 11…40
- ≥41

Layout creates relationships.

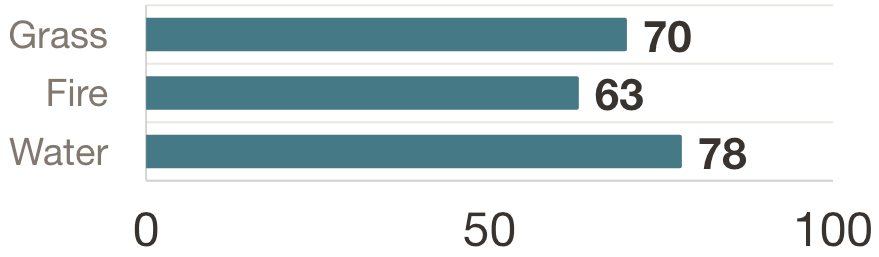
Same values. Different alignment.

SCATTERED

Attack

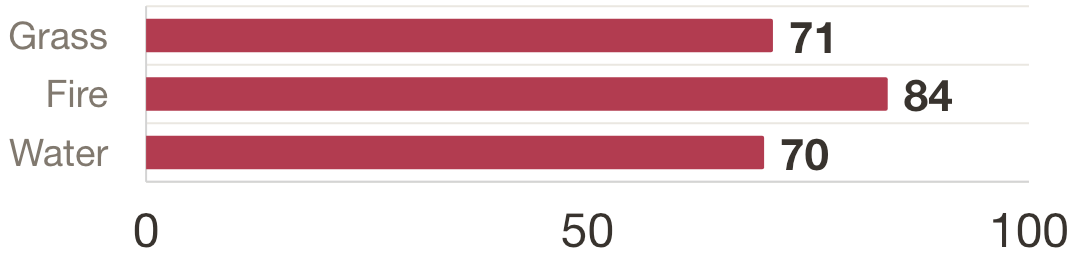


Defense

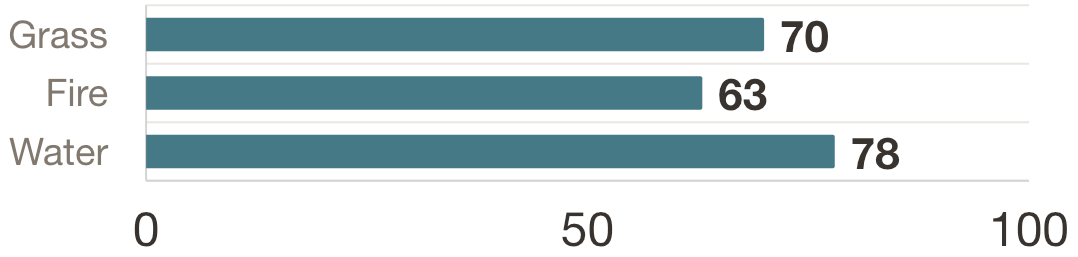


ALIGNED

Attack



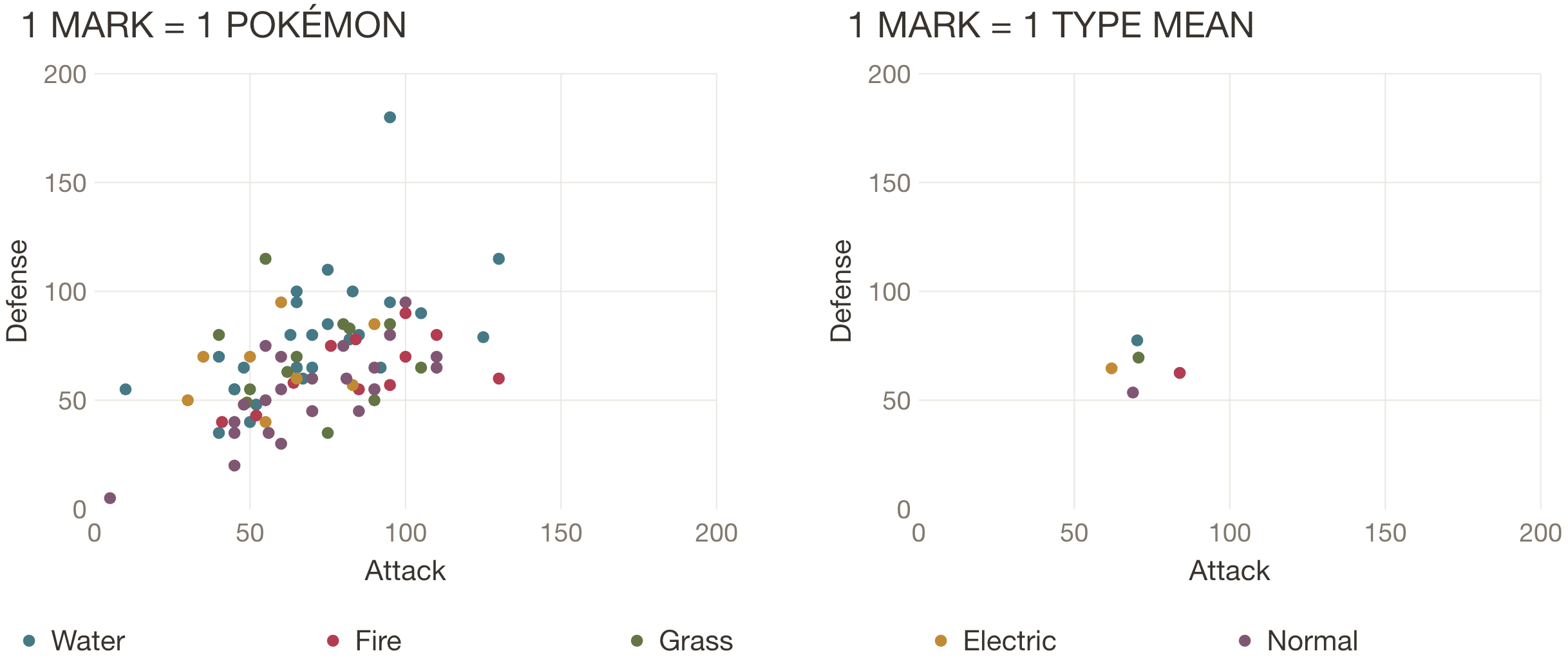
Defense



Mean base stats · primary type · rounded to whole numbers

What does one mark represent?

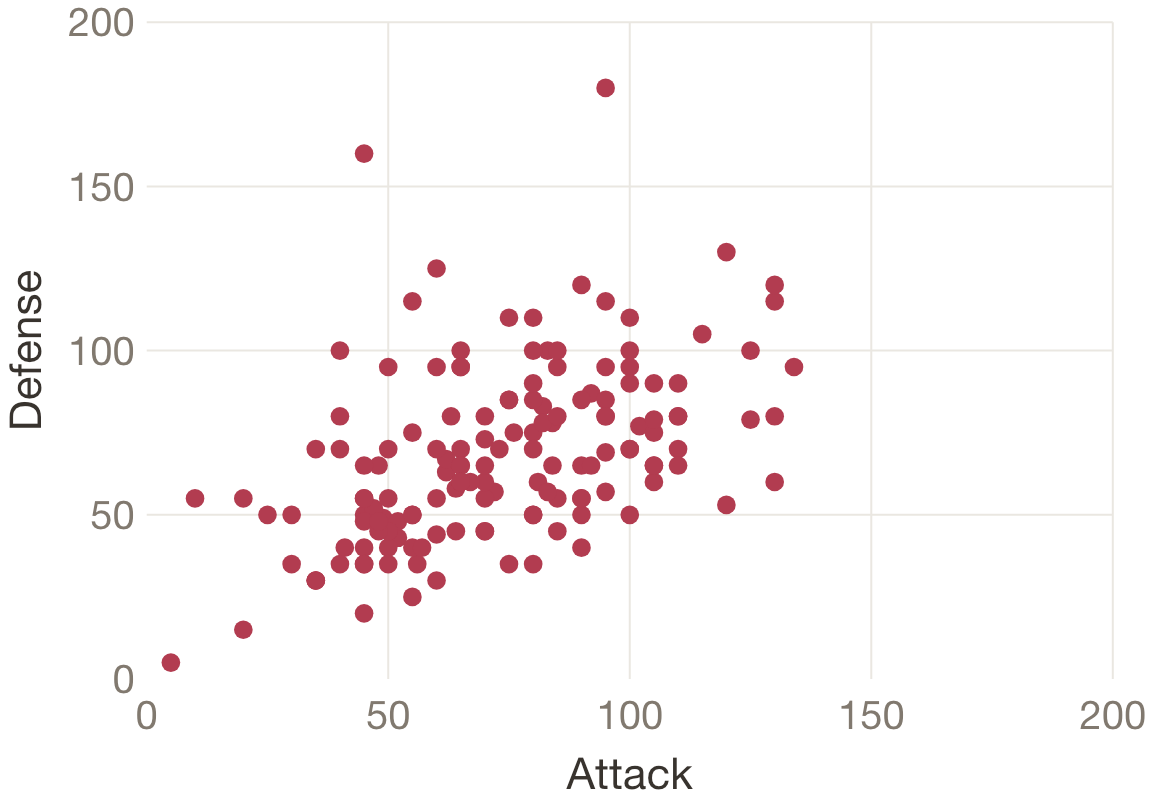
Aggregation changes the story.



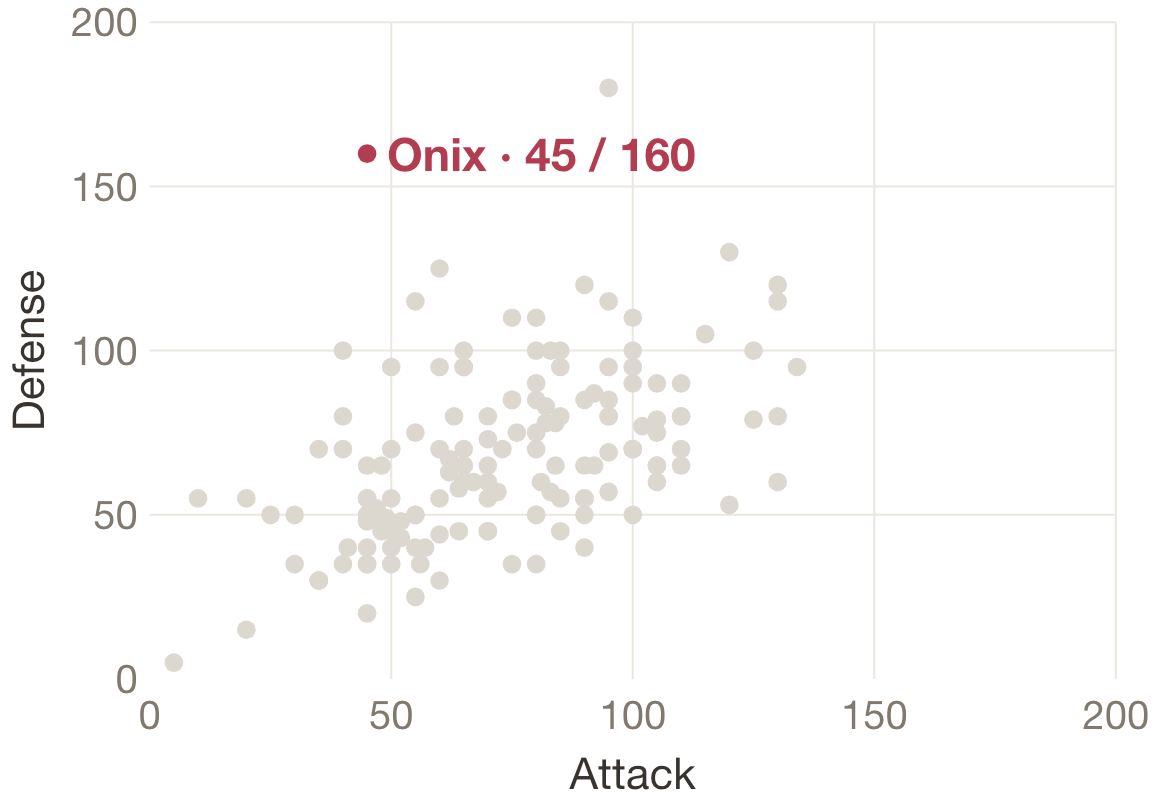
Help the reader find the point.

A title is a chance to say something.

Pokémon stats



Onix: high Defense, low Attack



ONE CLEAR CLAIM. ONE USEFUL LABEL.